QQ-W-414b
June 9, 1967
SUPERSEDING
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December 31, 1963

FEDERAL SPECIFICATION

WIRE, STEEL, CARBON: BOOKBINDER

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two types of bookbinder steel wire, wound on spools or cores.

1.2 Classification.

1.2.1 Types and finishes. Bookbinder steel wire shall be of the following types and finishes, as specified (see 6.2):

Type I - Flat, steel wire.
Type II - Round, steel wire.

Wire finishes:

A-Coppered.
B-Tinned.
C-Liquor finished.
D-Zinc coated.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.

Federal Standards:

Fed. Std. No. 66 - Steel: Chemical Composition and Hardenability.
Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-P-116 - Preservation, Methods of.

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Official Classification Committee:

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Ave. at 33rd Std., New York, N. Y., 10016.)

American Trucking Association:

National Motor Freight Classification Rules.

(Application for copies should be addressed to the American Trucking Association, 1616 P St., N.W., Washington, D. C., 20036.)

Fine and Specialty Wire Manufacturers' Association:

Standards Manual for Fine and Specialty Carbon Steel Wires.

(Application for copies should be addressed to the Fine and Specialty Wire Manufacturers' Association, 839 17th St., N.W., Washington, D. C.)

3. REQUIREMENTS

3.1 Description. The bookbinder steel wire shall be of uniform quality, temper, and finish and shall be furnished uniformly wound on spools or paper cores, as specified (see 6.2). The wire on each spool or core shall be in one continuous length without splicing. Wire ends shall be secured to prevent unwinding.

3.2 Material. The steel for the wire shall be made by the Bessemer, the open hearth, the electric furnace, or the basic oxygen furnace process. The steel shall conform to the chemical composition specified in table I.

TABLE I. Chemical compositions

Steel	Maximum values (percent)			
	Carbon	Manganese	Phosphorus	Sulfur
Bessemer	0.25	0.60	0.110	0.055
Open-hearth electric furnace, or B.O.F.	.25	.60	.045	.055
Tolerance, percent check analysis 1/	.02	.03	2/ .01 3/ .008	.008

1/ Fed. Std. No. 66.
2/ Applies to acid Bessemer steel.
3/ Applies to open-hearth steel.

3.3 Spools. Spools shall be made of wood, metal, plastic, fiber, or any acceptable combination of these materials. Unless otherwise specified (see 6.2), the spools shall be provided with a 5/8-inch-diameter spindle hole.

3.4 Cores. When paper cores are specified, the cores shall either fit the type and size of refillable spools as specified (see 6.2), or they shall be furnished with the quantity of refillable spools specified (see 6.2).

3.5 Identification marking. Each spool or core shall have attached a durable tag or label showing the manufacturer's name or trademark, wire type, diameter or flat wire dimensions, unit net weight, finish, and the letters "U.S.", in capitals.

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3.6 Finishes. The bookbinder wire shall be of the following finishes: A, coppered; B, tinned; C, liquor finished; D, zinc coated.

3.7 Type I, flat steel wire. The type I, flat steel wire shall conform to the Standards Manual of the Fine and Specialty Wire Manufacturer's Association, volume II. Size shall be as specified (see 6.2), and shall be as shown in table II. Tolerance on thickness shall be plus or minus 0.0005 inch.

TABLE II. Flat Wire

Size No.	Width (inch)	Tolerance on widths, minus (inch)	Thickness (inch)	Approximate length per pound (feet)
18 by 20	0.0475	0.0004	0.0348	206
19 by 21	.0410	.0004	.0317	248
19 by 21-1/2	.0410	.0004	.0301	290
20 by 22	.0348	.0004	.0286	305
20 by 23	.0348	.0004	.0258	350
20 by 24	.0348	.0004	.0230	415
20 by 25	.0348	.0004	.0204	478
21 by 25	.0317	.0003	.0204	525
22 by 26	.0286	.0003	.0181	644
23 by 30	.0258	.0003	.0140	750

3.8 Type II, round steel wire. The type II round steel wire shall conform to the Standards Manual of the Fine and Specialty Wire Manufacturer's Association. Size shall be as specified (see 6.2), and shall be as shown in table III.

TABLE III. Round wire

U.S. steel wire gage No.	Diameter (inch)	Tolerance on diameter (inch) plus or minus	Approximate length per pound (feet)
17	0.0540	0.0015	129
18	.0475	.0015	169
19	.0410	.0015	226
20	.0348	.0015	327
21	.0317	.0010	370
22	.0286	.0010	485
23	.0258	.0010	608
24	.0230	.0010	713
25	.0204	.0010	903
26	.0181	.0007	1.172
27	.0173	.0007	1.320
28	.0162	.0007	1.483
30	.0140	.0007	1.940

3.9 Workmanship. The finished, coated wire shall be smooth and free from kinks, waviness, bare or imperfectly coated spots, scratches, or other defects that will affect its use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of preparation for delivery (see 4.5).

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4.3 Quality conformance inspection.

4.3.1 Unit of product. For the purpose of inspection, a unit of product shall be one spool or core of wire.

4.3.2 Sampling.

4.3.2.1 For examination. Sampling for examination shall be in accordance with MIL-STD-105, inspection level S-3.

4.3.2.2 For tests. Samples of wire 12 inches in length shall be provided for each lot of 12 or less spools or cores. The samples shall be taken near the first, last, and intermediate units in the lot.

4.3.3 Examination. Samples selected in accordance with 4.3.2.1 shall be examined for the defects specified in 4.4.1. AQL shall be 6.5 percent defective.

4.3.4 Tests. Samples selected in accordance with 4.3.2.2 shall be tested as specified in 4.4.2. Failure of the tests shall be cause for rejection of the lot represented.

4.4 Inspection procedure.

4.4.1 Examination. The spool or core of wire shall be examined for the following defects:

101. Wire not uniform in finish.
102. Spools or cores not as specified.
103. Wire not in one continuous length.
104. Wire not uniformly wound on spools or cores.
105. Wire ends not properly secured to prevent unwinding.
106. Cores do not fit specified refillable spools.
107. Spindle hole not correct size.
108. Identification marking incorrect, illegible, or missing.
109. Decimal dimension of type I flat wire incorrect.
110. Decimal dimension of type II round wire incorrect.
111. Workmanship not as specified.

4.4.2 Tests. Ductility and tensile tests shall be conducted in accordance with the standards manual, volume II, as applicable. Nonconformance to the applicable standards manual shall constitute failure of these tests.

4.5 Inspection of preparation for delivery.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack of wire prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

112. Materials, methods, and containers not as specified for level A. Each incorrect material, method, or container shall constitute one defect.

113. Weight of contents exceed weight limit of shipping container for level A.

114. Strapping not zinc coated for level A.

115. Marking incomplete, incorrect, or illegible.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Each core or spool of wire shall be preserved and packaged in accordance with MIL-P-116, method III. The unit container shall conform to PPP-B-636, type CF, class weather-resistant, and shall be closed in accordance with the appendix to the box specification.

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5.1.2 Level C. Each core or spool of wire shall be packaged in a commercial container which will provide adequate protection against deterioration and damage from the supplier to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The cores or spools of wire of like description, packaged as specified in 5.1, shall be packed in snug-fitting boxes conforming to PPP-B-621, class 2, style optional, or PPP-B-601, overseas type, style optional. The gross weight of the boxes shall not exceed 200 pounds and the boxes shall be closed and strapped in accordance with the appendix to the applicable box specification. Strapping shall be zinc coated.

5.2.2 Level B. The wire, packaged as specified in 5.1, shall be packed in boxes conforming to PPP-B-636, class weather-resistant, grade V3c. Closure and strapping of the boxes shall be in accordance with the appendix to the box specification.

5.2.3 Level C. The wire, packaged as specified in 5.1, shall be packed in the quantities specified (see 6.2), to assure carrier acceptance and safe delivery to destination at lowest rates in containers complying with Uniform Freight Classification Rules or National Motor Freight Classification Rules.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special markings specified in the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special markings specified in the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The bookbinder steel wire is intended for use in manually-operated or automatically-operated bookbinding and paper stitching machines.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and finish required (see 1.2).
(c) Whether spools or paper cores shall be furnished (see 3.1).
(d) Diameter of spool spindle hole if other than as specified (see 3.3).
(e) Type and size of refillable spool required (see 3.4).
(f) Quantity of refillable spools required (see 3.4).
(g) Size of flat steel wire required (see 3.7).
(h) Size of round steel wire required (see 3.8).
(i) Selection of applicable levels of preservation, packaging, and packing required (see 5.1 and 5.2).
(j) Quantity to be packed (see 5.2.3).

6.3 This specification covers only the types, sizes, and finishes of the commodity as generally purchased by the Federal Government and is not intended to include all the types, sizes, and finishes which are commercially available.

MILITARY INTEREST:

Custodians:

Army - ME
Air Force - 69

Review activities:

Air Force - 69

User activity:

Army - MR

CIVIL AGENCIES:

GSA, HEW, VA, JUS, NBS, NSA

Preparing activity:

Army - ME

Code "N"

Other:

IS

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